



"Most beginners learn IOCs first...
But attackers fear IOAs more."



IOC vs IOA

The Complete Beginner's Guide
for SOC Analysts



Learn:



IOC Basics



IOA Basics



Real SOC
Examples



MITRE
ATT&CK
Mapping



Detection
Strategies



Investigation
Tips



Understand the difference.
Detect earlier. Respond faster. Stay ahead.

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1. WHAT ARE IOC AND IOA?



Imagine your house was robbed.
You arrive home and notice:

- Broken window
- Missing laptop
- Footprints on the floor



⇒ These signs are evidence that an attack already happened.

These signs are called
Indicators of Compromise (IOC).



Now imagine:

You see someone:

- Looking through your windows
- Checking door locks
- Walking around suspiciously



⇒ No crime happened yet.
But the behavior looks dangerous.

This is called
Indicator of Attack (IOA).

EASY DEFINITION



IOC = Evidence after attack
(What happened?)



IOA = Suspicious behavior
during attack
(What is happening right now?)

DAILY LIFE EXAMPLES

Situation	IOC (Evidence after attack)	IOA (Behavior during attack)
 House	Broken window, stolen items, footprints	Someone checking locks, looking through windows
 Bank	Money stolen, transaction done by attacker	Multiple failed login attempts, suspicious transfer attempts
 School	Exam paper leaked, computer infected	Student trying to access teacher PC, unusual USB usage

MINI EXERCISE

Think of one real-life example for IOC and one for IOA.

→ IOC Example: _____

→ IOA Example: _____



PRACTICAL TIP

Good SOC analysts look for IOAs early to stop attacks.
Don't wait for IOCs!





2. UNDERSTANDING IOC DEEPLY

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WHAT IS IOC?

IOC (Indicator of Compromise) is any piece of digital evidence that shows a system, network or user has been compromised by an attacker.



KEY IDEA

IOCs help us answer:
"What happened in the past?"

COMMON IOC TYPES

1. Malicious IP Address



IP addresses used by attackers for malicious activities.

Example:

185.199.110.153

2. Malicious Domain



Domain names used for phishing, malware hosting or C2.

Example:

bad-update.com

3. File Hash



Unique hash of a malicious file. Helps identify the exact file.

Example:

e3b0c44298fc...

4. Malware File



The actual malicious file detected on the system.

Example:

evil.exe

5. Registry Changes



Malicious or suspicious changes in Windows Registry.

Example:

HKCU\...\Run

6. Suspicious Process



Abnormal or malicious process running on the system.

Example:

powershell.exe

REAL-WORLD EXAMPLE



Your system makes an outbound connection to **185.199.110.153** which is known to host malware.

→ This IP address becomes an IOC.

IN SOC:

- ✓ Alert triggers for connection to known malicious IP.
- ✓ Analyst investigates the source, user and timeline.
- ✓ Actions are taken to contain and remove the threat.



PRACTICAL TIP

IOCs are great for identifying and investigating past attacks, cleaning infected systems and blocking known threats.

But attackers keep changing... that's why we also need IOAs!



KEY TAKEAWAY

- ✓ IOCs = Evidence
- ✓ Found AFTER the attack
- ✓ Helps in detection, investigation and response



MINI EXERCISE

List any 3 IOC examples that you have seen or can think of.

1. _____ 2. _____ 3. _____

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3. UNDERSTANDING IOA DEEPLY

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WHAT IS IOA?

IOA (Indicator of Attack) is any suspicious behavior that indicates an attacker is trying to compromise a system or the network.



KEY IDEA

IOA helps us answer: "What is happening right now?"



IOA focuses on **ACTIONS**, not evidence. It helps us detect attacks **early**, before real damage happens.

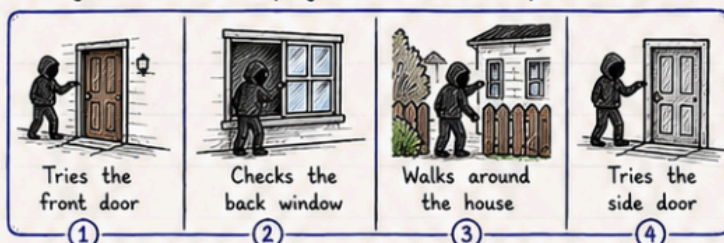
COMMON IOA EXAMPLES

- 1 Password spraying / brute force attempts
- 2 Multiple failed login attempts
- 3 Privilege escalation attempts
- 4 Lateral movement between systems
- 5 PowerShell or script abuse
- 6 Unusual data access or exfiltration attempts
- 7 Creating new user accounts
- 8 Disabling security tools or logs



DAILY LIFE EXAMPLE

Imagine someone trying to break into your house.



→ No theft yet. But the behavior is **SUSPICIOUS**. That's **IOA**!



PRACTICAL TIP

Attackers leave behaviors. IOAs help us catch them before they succeed. Detect early. Protect more.

IOA vs IOC (Quick Reminder)



IOA = Behavior

- During attack
- Predictive
- Helps in prevention
- Example: Multiple failed logins



IOC = Evidence

- After attack
- Reactive
- Helps in investigation
- Example: Malware hash found



IN SOC:

- ✓ IOAs trigger alerts in real time.
- ✓ Analyst investigates the behavior.
- ✓ If confirmed, we act before damage happens.
- ✓ Stop the attacker before they leave IOCs!

MINI EXERCISE

List any 3 IOA examples you can think of.

1. _____
2. _____
3. _____



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4. IOC TYPES EXPLAINED

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IOCs can come from many different places. They are digital evidence left by attackers after a system has been compromised.

IOC CATEGORIES

1 NETWORK IOC

Indicators found in network traffic and connections.



- Malicious IP Address
- Malicious Domain
- Malicious URL
- Unusual Port / Protocol

Example:

185.199.110.153
bad-site[.]com
http://malicious[.]xyz

2 HOST IOC

Indicators found on the endpoint or server.



- File Hash (MD5, SHA256)
- Malware / Suspicious File
- Registry Key Changes
- Suspicious Process / Service
- Log File Changes

Example:

SHA256: e3b0c44298fc...
C:\Users\Public\evil.exe
HKCU\...\Run\bad.exe

3 EMAIL IOC

Indicators found in suspicious emails and attachments.



- Suspicious Sender Address
- Phishing URL
- Malicious Attachment Hash
- Email Header Anomalies

Example:

attacker@bad-domain.com
http://phishing-site[.]com
SHA256: a1b2c3d4e5...

HOW IOCs ARE USED



Collect IOC

From threat intel,
logs, reports, etc.



Search / Match

Match IOC against
systems and traffic.



Alert Triggered

IOC found!
Generate alert.



Investigate & Respond

Analyst investigates
and responds.

EXAMPLES FROM DAILY LIFE

Situation	IOC Example
 House Break-in	Broken window, stolen items, fingerprints
 Bank Robbery	CCTV image of suspect, getaway car number
 Computer Virus	Malware file hash, malicious IP in logs
 Phishing Email	Malicious sender, phishing link, attachment hash

REMEMBER



IOCs help detect known threats.
But attackers keep changing...
That's why we also need IOAs!

PRACTICAL TIP

IOCs are your proof.
They help you answer:
"What already happened?"

Use IOC feeds, threat intelligence,
and SIEM tools to find them.



MINI EXERCISE

Match the following IOCs to their category:

- | | |
|-----------------------------|---------------------------|
| 1) 45.77.23.10 | 4) C:\evil.exe |
| 2) bad-update.com | 5) attacker@evil.com |
| 3) e7f6a1b2c3d4... (SHA256) | 6) http://malicious[.]xyz |

- | | | |
|----------|----------|----------|
| 1) _____ | 2) _____ | 3) _____ |
| 4) _____ | 5) _____ | 6) _____ |



5. IOA TYPES EXPLAINED

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IOAs are based on **ATTACKER BEHAVIOR**.

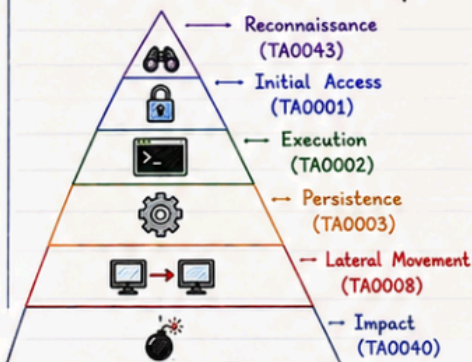
They show the steps an attacker takes during an attack lifecycle.

COMMON ATTACK BEHAVIORS (IOA TYPES)

1	2	3	4	5	6
RECONNAISSANCE	INITIAL ACCESS	EXECUTION	PERSISTENCE	LATERAL MOVEMENT	IMPACT
- Scanning ports - Enumerating users - Searching systems - Collecting info	- Brute force logins - Password spraying - Phishing attempts - Exploiting apps	- Running scripts - PowerShell abuse - Executing malware - Command execution	- Creating services - Scheduled tasks - Registry changes - Startup modifications	- RDP / SMB access - Pass-the-hash - Remote execution - Pivoting to others	- Data encryption - Data exfiltration - Deleting logs - Disrupting services
Goal: Gather information	Goal: Get into the system	Goal: Run malicious code	Goal: Stay on the system	Goal: Move to other systems	Goal: Achieve objective

IOA MAPPING WITH MITRE ATT&CK

MITRE ATT&CK is a global knowledge base of attacker behaviors and techniques.



Each IOA you detect usually maps to one or more **ATT&CK** techniques.

Mapping helps analysts understand Tactics, Techniques and Procedures (TTPs) used by attackers.

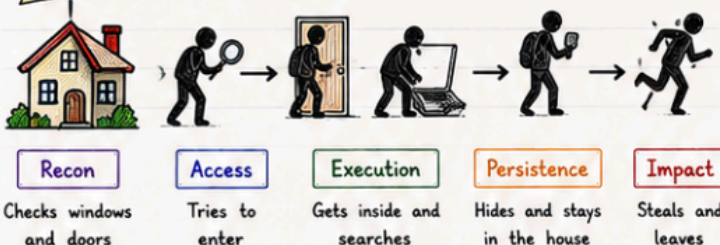
PRACTICAL TIP



Focus on behaviors, not just indicators.

- ✓ IOAs help detect attacks earlier.
- ✓ Monitor logs, commands, processes and network activity.
- ✓ Use EDR, SIEM and threat intelligence to catch attacker behaviors.
- ✓ The earlier you detect, the more damage you prevent!

DAILY LIFE EXAMPLE



MINI EXERCISE



Identify which IOA category the following behavior belongs to:

- Multiple failed login attempts _____
- PowerShell download cradle _____
- SMB access to multiple systems _____
- Creating a new scheduled task _____
- Scanning open ports on network _____



Remember: Attackers leave **BEHAVIORS** first, **EVIDENCE** later.



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6. IOC vs IOA COMPARISON

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Both **IOC** and **IOA** are important.
But they are different in many ways.
Let's **compare** them side-by-side!

Feature	IOC (Indicator of Compromise)	IOA (Indicator of Attack)
What it Detects	Evidence that an attack has already happened. after	Suspicious behavior that shows an attack is happening or about to happen.
When it Appears	After the damage is done.	During the attack (in real time).
Focus	Artifacts, files, IPs, hashes, logs, changes.	Attacker behavior, actions, techniques, patterns.
Speed of Detection	Usually slower .	Usually faster .
Helps With	Investigation & understanding what happened.	Prevention & stopping the attack early.
Prevention Power	Limited (damage may already be done).	Stronger (can stop attack before big damage).
Best Use	Post-incident analysis, forensics, threat hunting.	Real-time monitoring, alerting, threat detection.
Goal	Answer: "What happened?"	Answer: "What is happening right now?"

HOUSE SECURITY EXAMPLE

IOC (After the crime)	IOA (During the crime)
• Broken window • Missing TV • Footprints inside • Stolen items Evidence that someone already broke in.	• Someone checking windows • Trying the door lock • Looking around • Using tools nearby Suspicious behavior before the break-in happens.

PRACTICAL TIP



- ✓ Use IOAs to detect early.
- ✓ Use IOCs to confirm and investigate.
- ✓ Best SOC's use **BOTH** for stronger defense!



KEY TAKEAWAY

- ★ IOCs = Evidence of **past** attacks.
- ★ IOAs = Suspicious behavior in the **present**.
- ★ IOC tells you **where** you've been.
IOA helps you **where** you're going.
- ★ Use both to **detect**, **investigate** and **respond** better.



MINI EXERCISE



Match each scenario to IOC or IOA.

- 1) Multiple failed login attempts
- 2) Malware hash detected in file
- 3) PowerShell script execution
- 4) Connection to malicious IP
- 5) Admin privilege escalation attempt

IOC	IOA
☐	☐
☐	☐
☐	☐
☐	☐
☐	☐



Don't wait for **evidence**. Stop the attack while it's **happening**!



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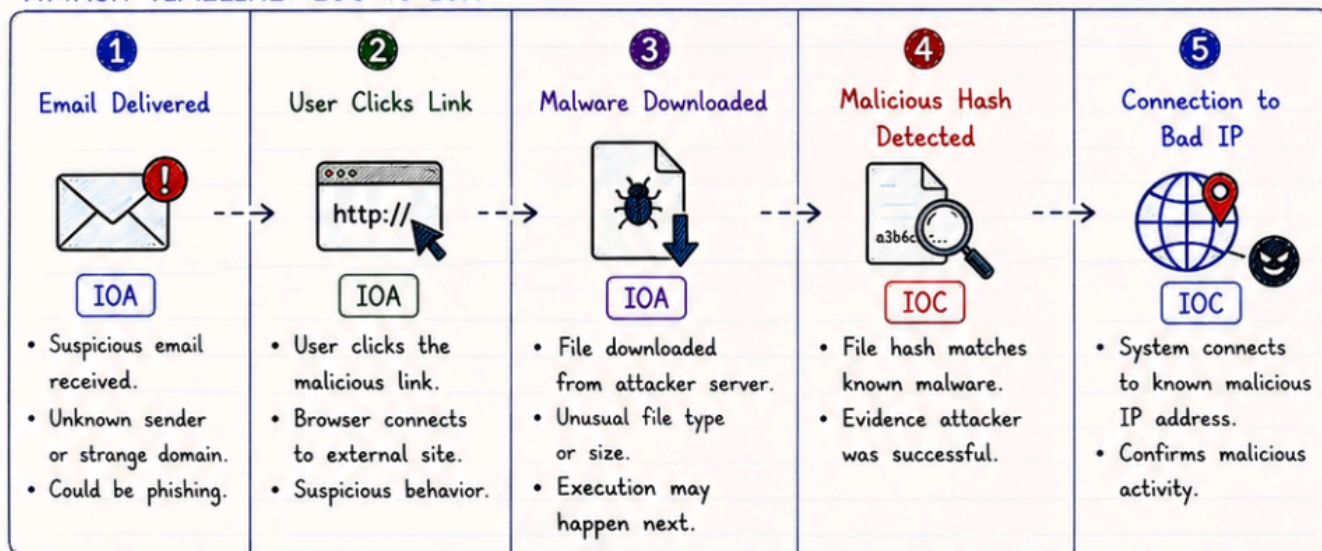
7. REAL SOC INVESTIGATION EXAMPLE

Let's see how IOC and IOA appear during a real-world attack.

Scenario: An employee receives a phishing email.



ATTACK TIMELINE: IOC vs IOA



LESSON LEARNED

- ☒ IOAs appear **FIRST**. They are warning signs.
- ☒ **IOCs** appear **LATER**. They are proof.
- ☒ If we detect IOAs early, we can **stop the attack** before damage happens!

DETAILED VIEW

Stage	What Happens	Type	Why it Matters
1 Email Delivered	Phishing email lands in inbox.	IOA	First suspicious event. Start investigating.
2 User Clicks Link	User clicks the link in the email.	IOA	Shows potential compromise attempt.
3 Malware Downloaded	Malware file downloaded to system.	IOA	Attacker trying to execute code.
4 Malicious Hash Detected	EDR/AV detects known bad hash.	IOC	Confirmation of malicious file.
5 Connection to Bad IP	System connects to C2 server.	IOC	Strong evidence of compromise.

PRACTICAL TIP



- ✓ Invest in tools that detect behavior (IOA).
- ✓ EDR, NDR, UEBA help catch early signs.
- ✓ Early detection = Less damage, lower cost!

MINI EXERCISE



Put (IOA) or (IOC) next to each event below.

- 1) File hash matches known malware _____
- 2) Multiple failed login attempts _____
- 3) Connection to 185.199.110.153 _____
- 4) PowerShell script execution _____
- 5) Suspicious email from unknown sender _____



Remember: Detect the **behavior** (IOA), find the **evidence** (IOC), **stop the attacker**!



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8. IOC & IOA IN SECURITY TOOLS



SOC analysts use **IOC** and **IOA** in many security tools every day to **detect**, **investigate** and **respond** to threats.

1 MICROSOFT DEFENDER

IOC (Indicators)

- Malware hash
- Malicious IP
- Known bad domain



IOA (Behaviors)

- PowerShell abuse
- Credential dumping
- Privilege escalation



2 CROWDSTRIKE

Focus: Behavior Detection

Examples:

- Lateral movement
- Persistence
- Suspicious scripts



3 SPLUNK

splunk>

Uses:

- IOC correlation
- IOA detection rules



4 QRADAR

Uses:

- Log correlation
- Offense creation
- IOC matching



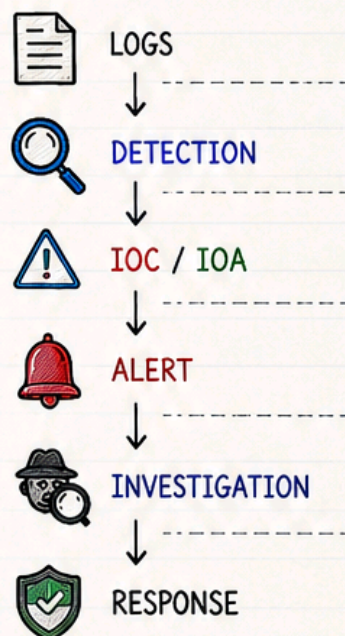
5 WAZUH

Uses:

- File Integrity Monitoring
- Threat Intelligence Matching
- Behavioral Detection



SOC WORKFLOW



HOW IOC & IOA ARE USED TOGETHER

IOC = Known bad (Evidence)



We look for IOCs to find proof that an attack happened.



IOA = Suspicious behavior



We look for IOAs to catch attacks before damage happens.



IOC confirms the attack. IOA helps us stop the attack early.



PRACTICAL TIP

SOC analysts **don't wait** for alerts.

They actively search for IOAs and check IOCs to keep the environment safe.



Remember: Great defenders look for **BEHAVIOR** first, then confirm with **EVIDENCE**.





9. IOC & IOA IN A REAL RANSOMWARE ATTACK

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Let's walk through a real ransomware attack and see where IOAs (behaviors) and IOCs (evidence) show up.

1	ATTACKER SCANS NETWORK Attacker scans open ports and services.	→	IOA Reconnaissance behavior.	<u>Examples</u> • Port scanning • Network discovery	
2	PASSWORD SPRAYING Attacker tries many common passwords on many accounts.	→	IOA Brute force / Credential access behavior.	<u>Examples</u> • Multiple failed logins • From different IPs	
3	POWERSHELL ABUSE Attacker runs PowerShell commands to execute scripts.	→	IOA Execution & living off the land behavior.	<u>Examples</u> • PowerShell execution • Encoded commands	
4	PRIVILEGE ESCALATION Attacker tries to gain higher privileges.	→	IOA Privilege escalation behavior.	<u>Examples</u> • Token impersonation • Exploiting vulnerabilities	
5	RANSOMWARE EXECUTED Attacker executes the ransomware payload.	→	IOA Malicious execution behavior.	<u>Examples</u> • Suspicious process • Mass file encryption	
6	KNOWN MALWARE HASH FOUND File hash matches known malware in threat intel.	→	IOC Evidence of a known threat.	<u>Examples</u> • File hash match (SHA256)	
7	CONNECTION TO C2 IP System connects to attacker's command & control server.	→	IOC Evidence of malicious network activity.	<u>Examples</u> • Connection to known bad IP	
8	RANSOM NOTE APPEARS Ransom note is dropped on the system.	→	IOC Evidence of impact & final stage.	<u>Examples</u> • Ransom note file • .txt / .html file	



LESSON LEARNED

IOAs appear **early**. They are our warning signs.
IOCs appear **after** damage or during the final stages.
The earlier we detect IOAs, the fewer **IOCs** we will see later!



DAILY LIFE EXAMPLE

Smoke



= **IOA**
(Early Warning)

Fire Damage



= **IOC**
(After Damage)



KEY TAKEAWAY

- ✓ Detecting IOAs early helps stop the attack before it causes major damage.
- ✓ IOCs help us confirm what happened and remove the threat completely.
- ✓ Great defenders focus on **behavior** first, **evidence** second.



Remember: Early detection (**IOA**) = Less damage, Less cost, More protection!

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10. SOC ANALYST CHEAT SHEET

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Your ultimate quick reference for **IOC** vs **IOA**.

IOC (INDICATOR OF COMPROMISE)



IOA (INDICATOR OF ATTACK)



IOC = Evidence



IOA = Behavior



IOC = Past



IOA = Present



IOC = Investigation



IOA = Prevention



IOC = What happened?



IOA = What is happening?

QUICK EXAMPLES

IOC (Evidence of past attack)



Malware Hash

Known malicious file hash



Known Bad IP

IP address from threat intel



Malicious Domain

Domain linked to attacks



Malicious Email

Email with malicious attachment



Malicious File

File identified as malicious

IOA (Suspicious behavior)



Multiple Failed Logins

Many failed login attempts



PowerShell Abuse

Suspicious PowerShell commands



Privilege Escalation

User gains higher privileges



Lateral Movement

Accessing systems laterally



Unusual File Download

Large or unusual file downloads

INTERVIEW QUESTIONS ?

Q1. What is the difference between IOC and IOA?

A: IOC is evidence of a past attack.

IOA is suspicious behavior indicating an ongoing attack.

Q2. Which helps detect attacks earlier?

A: IOA helps detect attacks earlier.

Q3. Give 3 examples of IOC.

A: 1) Malware Hash 2) Malicious IP 3) Malicious Domain

Q4. Give 3 examples of IOA.

A: 1) Multiple Failed Logins 2) PowerShell Abuse
3) Privilege Escalation

MEMORY TRICK

Smoke

= IOA



(Warning Sign)

Ashes

= IOC



(Proof It Happened)



Think: Smoke comes before the fire.
IOA comes before IOC.

FINAL TAKEAWAY



Elite SOC Analysts
don't wait for IOCs.
They hunt IOAs
before damage happens.



IOA = Detect Early
Stop the Attack



IOC = Confirm & Respond
Remove the Threat



Be proactive. Be observant. Be the defender who stops attacks early!



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11. READY TO LEVEL UP YOUR BLUE TEAM SKILLS?



You've learned IOC vs IOA. Now it's time to go deeper, build real skills and become an elite defender.

Blueteamers.io is your roadmap to a successful SOC career!

WHAT YOU'LL LEARN AT BLUETEAMERS.IO

1. SOC ANALYST NOTES



- Complete SOC concepts
- Easy to understand notes
- Perfect for beginners
- Interview ready content

2. INCIDENT RESPONSE



- Step-by-step IR process
- Real incident examples
- Playbooks & checklists
- Become IR ready

3. THREAT HUNTING



- Hunt like a pro
- Hypotheses & methods
- Real hunting scenarios
- Find threats early

4. SIEM TUTORIALS



- Splunk, QRadar, Wazuh
- Queries & searches
- Dashboards & alerts
- Hands-on labs

5. DETECTION ENGINEERING



- Create strong detections
- Sigma & rules
- Improve SOC visibility
- Reduce false positives

6. CAREER ROADMAPS



- Step-by-step career paths
- Skills & tools checklist
- Certification guidance
- Land your dream job

7. BEGINNER FRIENDLY GUIDES



- Start from zero
- Simple explanations
- Practical examples
- Build confidence daily

8. LABS & PRACTICE



- Hands-on labs
- Real-world practice
- Attack & defense labs
- Learn by doing

WHY CHOOSE BLUETEAMERS.IO?



BEGINNER FRIENDLY

No prior experience?
No problem!



PRACTICAL & HANDS-ON

Learn by doing,
not just reading.



REAL WORLD FOCUSED

Skills that SOC's
actually use.



CAREER GROWTH

Get job ready.
Stay ahead.



TRUSTED BY THOUSANDS

Join a growing
community of defenders!



DON'T JUST LEARN ALERTS.
LEARN HOW ATTACKERS OPERATE.
LEARN HOW DEFENDERS WIN.



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DETECT EARLY. INVESTIGATE SMART. RESPOND FAST. STAY AHEAD!

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